

Programming Assignment 2 Part 2 Report

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1. Algorithm Design

I represented the raw Metro graph using Adjacency Lists. To handle the two types of edges, I maintained separate lists: graphSubway for weighted subway connections and graphTransfer for -1 walking connections. I defined a "Hub Station" as a connected component of vertices linked by -1 edges. At the same time, I used Breadth-First Search (BFS) to traverse these -1 edges and group vertices. Further, I used an array vertexToHubId to map every original vertex ID to a simplified Hub ID.

The vertices of the new graph are the identified Hubs. To find edges (Segments), I performed a Depth-First Search (DFS) starting from every Hub vertex. The DFS follows subway edges through non-hub stations until it encounters another Hub vertex. The total weight of the path becomes the weight of the edge in the simplified graph. Based on the assignment requirement, I reused the Kruskal's Algorithm with the Partition ADT (Linked List implementation) from Part A to solve for the minimum cost.

2. Testing & Status

The program was tested using the provided metro.txt via input redirection. After running the program, it correctly identified 58 Hub Stations. The MST cost and lists of the segments were successfully calculated. The output was as follows:

```
C:\Windows\System32\cmd.exe
D:\Ottawa\3rd term\CSI\Assignment\Programming Assignment 2>javac ParisMetro.java
D:\Ottawa\3rd term\CSI\Assignment\Programming Assignment 2>java ParisMetro < metro.txt
Paris Metro Graph has 376 vertices and 933 edges.
Hub Stations = [ Arts et Métiers, Barbès Rochechouart, Bastille, Belleville, ... ]
Number of Hub Stations = 58 (total Hub Vertices = 138)
Number of Possible Segments = 245
Total Cost = $4464
Segments to Buy:
1( Jaurès - Stalingrad ) - $27
2( Michel Ange Auteuil - Michel Ange Molitor ) - $28
3( Denfert Rochereau - Raspail ) - $37
4( Havre Caumartin - Saint-Lazare ) - $40
5( Bonne Nouvelle - Rue Montmartre, Grands Boulevards ) - $41
6( Concorde - Madeleine ) - $41
7( Gare de Lyon - Reuilly Diderot ) - $41
8( Chaussée d'Antin, La Fayette - Havre Caumartin ) - $45
9( Barbès Rochechouart - Gare du Nord ) - $46
10( Jaurès - Louis Blanc ) - $46
11( Richelieu Drouot - Rue Montmartre, Grands Boulevards ) - $46
12( Palais Royal, Musée du Louvre - Pyramides ) - $49
13( Bonne Nouvelle - Strasbourg Saint-Denis ) - $52
14( Champs Élysées, Clémenceau - Concorde ) - $54
15( Montparnasse Bienvenue - Pasteur ) - $54
16( Arts et Métiers - Réaumur Sébastopol ) - $58
17( Gare d'Austerlitz - Jussieu ) - $59
18( Gare de l'Est - Gare du Nord ) - $59
19( Champs Élysées, Clémenceau - Franklin D. Roosevelt ) - $62
20( Champs Élysées, Clémenceau - Invalides ) - $62
21( Chaussée d'Antin, La Fayette - Richelieu Drouot ) - $63
22( Duroc - Sévres Babylone ) - $64
23( Nation - Reuilly Diderot ) - $64
24( Réaumur Sébastopol - Strasbourg Saint-Denis ) - $67
25( Havre Caumartin - Opéra ) - $68
26( Champs Élysées, Clémenceau - Miromesnil ) - $69
27( Gambetta - Père Lachaise ) - $70
28( Madeleine - Pyramides ) - $71
29( Saint-Lazare - Villiers ) - $74
30( Arts et Métiers - République ) - $75
31( Daumesnil - Reuilly Diderot ) - $77
32( Place de Clichy - Saint-Lazare ) - $77
33( Montparnasse Bienvenue - Raspail ) - $79
34( Bercy - Daumesnil ) - $80
35( Oberkampf - République ) - $80
36( Opéra - Pyramides ) - $80
37( Place des Fêtes - Porte des Lilas ) - $82
38( Barbès Rochechouart - Marcadet Poissonniers ) - $85
```

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No known bugs were found in code. The solution handles parallel segments (multiple routes between hubs) correctly because Kruskal's algorithm automatically selects the edge with the lowest weight.